

Archi associati: riepilogo

Prof. Vincenzo Resta

angoli supplementari

$$\cos(\pi - \alpha) = -\cos \alpha$$

$$\sin(\pi - \alpha) = \sin \alpha$$

$$\operatorname{tg}(\pi - \alpha) = -\operatorname{tg} \alpha$$

$$\operatorname{ctg}(\pi - \alpha) = -\operatorname{ctg} \alpha$$

angoli che differiscono di π

$$\cos(\pi + \alpha) = -\cos \alpha$$

$$\sin(\pi + \alpha) = -\sin \alpha$$

$$\operatorname{tg}(\pi + \alpha) = \operatorname{tg} \alpha$$

$$\operatorname{ctg}(\pi + \alpha) = \operatorname{ctg} \alpha$$

angoli complementari

$$\cos\left(\frac{\pi}{2} - \alpha\right) = \sin \alpha$$

$$\sin\left(\frac{\pi}{2} - \alpha\right) = \cos \alpha$$

$$\operatorname{tg}\left(\frac{\pi}{2} - \alpha\right) = \operatorname{ctg} \alpha$$

$$\operatorname{ctg}\left(\frac{\pi}{2} - \alpha\right) = \operatorname{tg} \alpha$$

angoli che differiscono di $\frac{\pi}{2}$

$$\cos\left(\frac{\pi}{2} + \alpha\right) = -\sin \alpha$$

$$\sin\left(\frac{\pi}{2} + \alpha\right) = \cos \alpha$$

$$\operatorname{tg}\left(\frac{\pi}{2} + \alpha\right) = -\operatorname{ctg} \alpha$$

$$\operatorname{ctg}\left(\frac{\pi}{2} + \alpha\right) = -\operatorname{tg} \alpha$$

angoli esplementari – angoli opposti

$$\cos(2\pi - \alpha) = \cos(-\alpha) = \cos \alpha$$

$$\sin(2\pi - \alpha) = \sin(-\alpha) = -\sin \alpha$$

$$\operatorname{tg}(2\pi - \alpha) = \operatorname{tg}(-\alpha) = -\operatorname{tg} \alpha$$

$$\operatorname{ctg}(2\pi - \alpha) = \operatorname{ctg}(-\alpha) = -\operatorname{ctg} \alpha$$

angoli la cui somma è $\frac{3\pi}{2}$

$$\cos\left(\frac{3}{2}\pi - \alpha\right) = -\sin \alpha$$

$$\sin\left(\frac{3}{2}\pi - \alpha\right) = -\cos \alpha$$

$$\operatorname{tg}\left(\frac{3}{2}\pi - \alpha\right) = \operatorname{ctg} \alpha$$

$$\operatorname{ctg}\left(\frac{3}{2}\pi - \alpha\right) = \operatorname{tg} \alpha$$

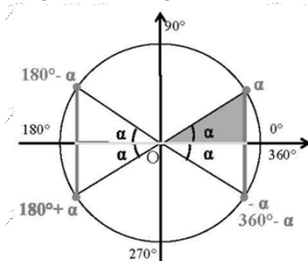
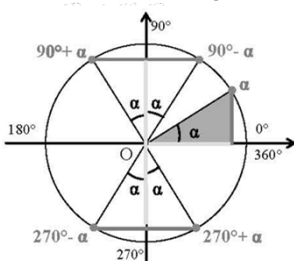
angoli che differiscono di $\frac{3\pi}{2}$

$$\cos\left(\frac{3}{2}\pi + \alpha\right) = \sin \alpha$$

$$\sin\left(\frac{3}{2}\pi + \alpha\right) = -\cos \alpha$$

$$\operatorname{tg}\left(\frac{3}{2}\pi + \alpha\right) = -\operatorname{ctg} \alpha$$

$$\operatorname{ctg}\left(\frac{3}{2}\pi + \alpha\right) = -\operatorname{tg} \alpha$$



30

Formule goniometriche

Prof. Vincenzo Resta

Formule di addizione

$$\sin(\alpha \pm \beta) = \sin \alpha \cdot \cos \beta \pm \cos \alpha \cdot \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cdot \cos \beta \mp \sin \alpha \cdot \sin \beta$$

$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \cdot \tan \beta}$$

Formule di duplicazione

$$\sin 2\alpha = 2 \sin \alpha \cdot \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

Formule di bisezione

$$\sin \frac{\alpha}{2} = \pm \sqrt{\frac{1 - \cos \alpha}{2}}$$

$$\cos \frac{\alpha}{2} = \pm \sqrt{\frac{1 + \cos \alpha}{2}}$$

$$\tan \frac{\alpha}{2} = \pm \sqrt{\frac{1 - \cos \alpha}{1 + \cos \alpha}}$$

Formule di Werner

$$\sin \alpha \cdot \sin \beta = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)]$$

$$\cos \alpha \cdot \cos \beta = \frac{1}{2} [\cos(\alpha + \beta) + \cos(\alpha - \beta)]$$

$$\sin \alpha \cdot \cos \beta = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)]$$

Formule di prostaferesi

$$\sin \alpha + \sin \beta = 2 \sin \frac{\alpha + \beta}{2} \cdot \cos \frac{\alpha - \beta}{2}$$

$$\sin \alpha - \sin \beta = 2 \cos \frac{\alpha + \beta}{2} \cdot \sin \frac{\alpha - \beta}{2}$$

$$\cos \alpha + \cos \beta = 2 \cos \frac{\alpha + \beta}{2} \cdot \cos \frac{\alpha - \beta}{2}$$

$$\cos \alpha - \cos \beta = -2 \sin \frac{\alpha + \beta}{2} \cdot \sin \frac{\alpha - \beta}{2}$$

Formule parametriche

$$\sin \alpha = \frac{2 \tan \frac{\alpha}{2}}{1 + \tan^2 \frac{\alpha}{2}}$$

$$\cos \alpha = \frac{1 - \tan^2 \frac{\alpha}{2}}{1 + \tan^2 \frac{\alpha}{2}}$$

51